

ASSIGNMENT SUBMISSION

Assignment Title: Development of QuickBasket AI: A Smart Inventory and Sales Management System Using Artificial Intelligence

Submitted By:

Name: Ayush Kumar Singh

Course: BBA (Artificial Intelligence & Data Science)

Semester: 3rd

College Name:

Kristu Jayanti (Deemed To Be University)

Submitted To:

FACE Prep

Domain:

Business Process Improvement (Artificial Intelligence & Machine Learning)

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PROJECT REPORT

QuickBasket AI

Smart Inventory & Sales Management System with AI-Based Business Insights

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Project Overview

QuickBasket AI is a web-based Smart Inventory and Sales Management System developed to help small grocery stores and retail businesses efficiently manage inventory, sales, customers, suppliers, and business reports. The application integrates Artificial Intelligence to provide sales forecasting, demand prediction, inventory analysis, and business insights, enabling smarter decision-making.

The system is built using Python Flask, SQLAlchemy, Bootstrap, PostgreSQL, and Machine Learning libraries. It offers secure user authentication, automated inventory tracking, invoice generation, detailed reporting, and cloud deployment.

Problem Statement

Many small retail shops still manage inventory and sales manually using notebooks or spreadsheets. These methods often lead to:

- Stock shortages
- Overstocking
- Human errors
- Poor sales analysis
- Time-consuming record management
- Lack of business insights

There is a need for an intelligent system that automates daily business operations while providing predictive analytics.

Objectives

- Develop a secure web-based inventory management system.
- Automate product and stock management.
- Record and manage daily sales transactions.
- Maintain customer and supplier information.

- Generate business reports.
- Integrate Artificial Intelligence for sales prediction.
- Deploy the application on the cloud.
- Ensure data persistence using PostgreSQL.

Proposed Solution

QuickBasket AI provides an integrated platform where shop owners can:

- Manage products and inventory
- Track sales transactions
- Generate invoices
- Monitor stock levels
- Manage suppliers and customers
- View graphical business reports
- Receive AI-generated business insights
- Predict future sales using Machine Learning

Technologies Used

Category	Technology
Programming Language	Python
Backend Framework	Flask
Frontend	HTML, CSS, Bootstrap 5, JavaScript
Database	SQLite (Development), PostgreSQL (Production)
ORM	SQLAlchemy
Authentication	Flask-Login

Category	Technology
Database Migration	Flask-Migrate
Machine Learning	Scikit-learn
Data Analysis	Pandas, NumPy
Visualization	Matplotlib
Excel Export	OpenPyXL
PDF Generation	ReportLab
Deployment	Render
Version Control	Git & GitHub

System Modules

1. User Authentication

Features:

- User Registration
- Secure Login
- Password Hashing
- Session Management
- Logout

2. Dashboard

Displays:

- Total Products
- Total Sales
- Revenue
- Low Stock Alerts

- Business Summary
- AI Insights

3. Inventory Management

Features:

- Add Product
- Edit Product
- Delete Product
- Product Categories
- Barcode Support
- Buying Price
- Selling Price
- Profit Calculation
- Stock Quantity
- Expiry Date
- Supplier Information
- Low Stock Monitoring

4. Sales Management

Features:

- Billing System
- Shopping Cart
- Invoice Generation
- GST Calculation
- Discount Management
- Automatic Stock Deduction

- Payment Methods
- Sales History

5. Customer Management

Stores:

- Customer Name
- Phone Number
- Purchase History

6. Supplier Management

Stores:

- Supplier Details
- Contact Information
- Product Supply Records

7. Reports Module

Generates:

- Sales Report
- Inventory Report
- Revenue Report
- Low Stock Report
- Monthly Sales Charts
- Excel Export
- CSV Export

8. AI Insights Module

Provides:

- Sales Prediction
- Demand Forecasting
- Business Recommendations
- Revenue Forecast
- Stock Risk Prediction
- Top Selling Products

Database Design

Main Tables

- Users
- Products
- Sales
- Sale Items
- Customers
- Suppliers

Relationships

- One User → Multiple Sales
- One Sale → Multiple Sale Items
- One Product → Multiple Sale Items
- One Supplier → Multiple Products

Artificial Intelligence Implementation

The AI module uses historical sales data to generate:

- Future Revenue Prediction
- Sales Forecast Charts

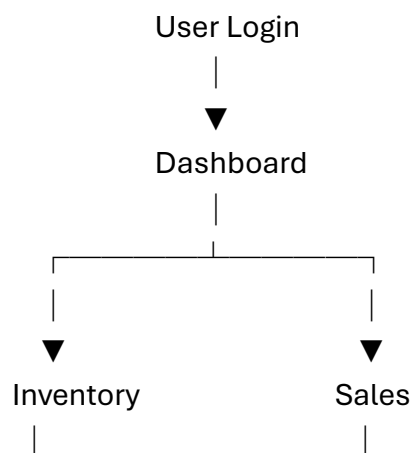
- Demand Analysis
- Inventory Risk Alerts
- Business Performance Insights

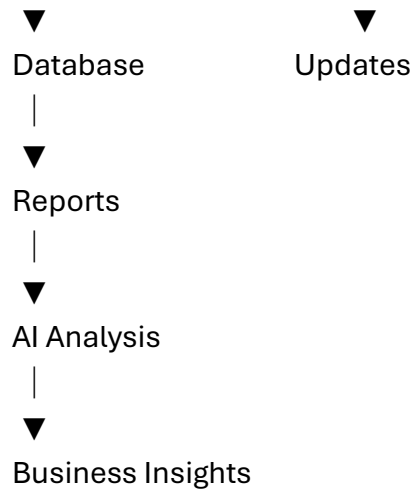
Machine Learning algorithms analyze previous sales records and estimate future business trends.

Key Features

- Secure Authentication
- Inventory Tracking
- Sales Management
- Invoice Generation
- AI-Based Predictions
- Business Dashboard
- Export Reports
- Responsive Design
- PostgreSQL Support
- Cloud Deployment

Project Workflow





Advantages

- Reduces manual work
- Improves inventory accuracy
- Provides AI-powered insights
- Generates professional reports
- Enhances business decisions
- Secure cloud deployment
- Scalable architecture

Limitations

- AI predictions depend on available historical sales data.
- Barcode scanning requires compatible hardware.
- Email notifications are not yet integrated.
- Multi-store management is not implemented.

Future Enhancements

- Mobile Application

- QR Code Billing
- WhatsApp Invoice Sharing
- Email Notifications
- Role-Based Access Control
- Barcode Scanner Integration
- Multi-Store Management
- Cloud Backup
- AI Chatbot Assistant
- Voice-Based Inventory Search

Testing

Module	Status
Login	Passed
Registration	Passed
Inventory	Passed
Sales	Passed
Customers	Passed
Suppliers	Passed
Reports	Passed
AI Insights	Passed
Deployment	Passed

Deployment

The application is deployed on **Render** using:

- Gunicorn

- PostgreSQL
- GitHub Integration
- Automatic Deployment

Learning Outcomes

During this project, the following skills were developed:

- Flask Web Development
- Database Design
- SQLAlchemy ORM
- Machine Learning Integration
- Data Visualization
- Cloud Deployment
- Version Control using Git
- RESTful Application Development
- Debugging and Testing
- Problem Solving

Conclusion

QuickBasket AI successfully demonstrates how Artificial Intelligence can enhance inventory and sales management for small retail businesses. The application automates essential business operations while providing predictive analytics, graphical reports, and intelligent recommendations. With its modular architecture, secure authentication, cloud deployment, and AI-powered insights, the system offers an efficient and scalable solution for modern retail management.